

Pei-Han Hsu

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HIGHLIGHT

- ETL/ELT pipeline design and schema modeling for **large-scale analytics and ML datasets**.
- Cloud-native data engineering with **AWS, GCP, Spark, Hadoop**.
- Production-ready workflows in **Python, SQL, Scala**, optimized for speed and scalability.

EDUCATION

Northeastern University

M.S in Information Systems | GPA: 3.7/4.0

Boston, MA

Sep 2023 - Apr 2025

Soochow University

B.B.A in Computer Science & Information Management | GPA: 3.9/4.0 (Dean's List)

Taipei, Taiwan

Sep 2018 - Jan 2022

SKILLS

Data Engineering & Cloud: Python, SQL, Scala, Java, ETL/ELT, dbt, Airflow, Snowflake, BigQuery, Redshift, MongoDB, Kafka, Flink, AWS, GCP, Azure, Docker/Kubernetes, CI/CD

Machine Learning & Analytics: scikit-learn, TensorFlow, PyTorch, Feature Engineering, Model Deployment (FastAPI, Flask), Data Quality Validation, Jupyter

Visualization & Reporting: Tableau, Looker, Dashboard Design, Metric Tracking & Monitoring

WORK EXPERIENCE

Data Analytics Engineer | HydraEarth Network | New Brunswick, NJ

Jul 2025 - Present

- Engineered ETL pipelines using **Python (Pandas, PySpark)** and **SQL** to process and transform **10M+ daily records**, streamlining analytics workflows and boosting data throughput by **40%**.
- Designed cloud-integrated infrastructure by connecting **Looker Studio** and **Tableau** to **AWS S3**, enabling **real-time monitoring dashboards** and cutting reporting latency by **70%**.
- Automated production-ready workflows with **dbt** and orchestrated pipelines in **Airflow**, reducing end-to-end data latency by **50%** while improving analytical query performance by **35%**.

Data Infrastructure Engineer | Multimedia & Mobile Application Lab | Hybrid

Sep 2020 - Dec 2024

- Designed relational and NoSQL architectures to manage **multimodal research datasets** (text, image, sensor), enabling seamless integration with **ML experiments** and supporting **5+ academic projects**.
- Implemented distributed data platforms with **Hadoop** and **Spark** on **AWS/GCP**, scaling compute resources to process **2TB+ weekly workloads** with fault-tolerance for long-running batch jobs.
- Developed **parallelized algorithms** in **Python, Scala, and SQL** to accelerate feature engineering and model training, cutting **pipeline execution time by 60%** and improving cross-cluster utilization.

Software Development Engineer | Tera ASPer Digital Services Co. | Hybrid

Jan - Sep 2021

- Built responsive user interfaces with **HTML, CSS, and JavaScript**, improving mobile compatibility and reducing average page load time by **30%**.
- Developed RESTful APIs in **Java**, optimizing request handling to lower response latency by **25%** and strengthen cross-system data exchange.
- Designed normalized **MySQL schemas** supporting **100k+ records** and implemented indexing/optimization for **45% faster queries**; contributed to early-stage ML feature pipelines by structuring clean, query-ready data.

PROJECTS

Medical Adherence Chatbot for Elderly Users | Azure, Rasa, Swift, Python

- Built an iOS interface in **Swift** to capture medication inputs and manage reminders, enabling seamless **real-time tracking** across **200+ user sessions**.
- Deployed an NLP-driven chatbot with **Rasa on Azure**, delivering dynamic conversation flows and sustaining concurrent interactions with **<300ms response latency**.
- Designed secure, normalized schemas and structured **user log datasets** to support downstream **ML models** for predicting medication adherence patterns.